BREAST CANCER DETECTION

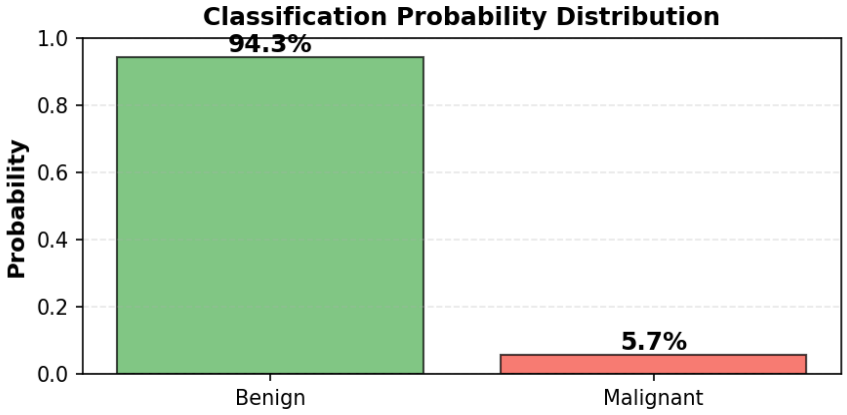
AI-Assisted Histopathology Analysis Report

Report ID:	BCR-20260216_192946
Generated:	2026-02-16 19:29:46
Patient ID:	N/A
Image File:	tmpc5b8jlfx.png

DIAGNOSTIC RESULTS

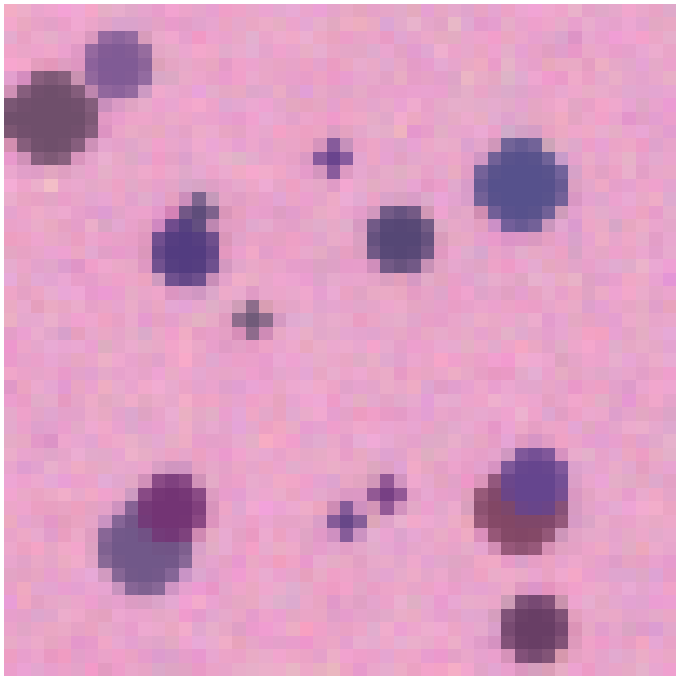
Classification:	**BENIGN**
Confidence Score:	94.3%
Benign Probability:	94.3%
Malignant Probability:	5.7%

Probability Distribution



HISTOPATHOLOGY IMAGE ANALYSIS

Original Histopathology Image



CLINICAL RECOMMENDATIONS

Routine screening after 12 months. Continue regular breast health monitoring. Maintain healthy lifestyle and attend scheduled screenings.

Risk Level:	**LOW**
Follow-up Timeline:	12 months (routine screening)
Specialist Referral:	Primary Care Physician (routine follow-up)

IMPORTANT MEDICAL DISCLAIMER

This report is generated by an AI-assisted diagnostic system and is intended for decision support purposes only.

Key Points:
• This analysis must be validated by qualified medical professionals
• Do NOT make treatment decisions based solely on this AI analysis
• Always consult with board-certified pathologists and oncologists
• This system is not a substitute for professional medical judgment
• False positives and false negatives can occur
• Clinical correlation and additional diagnostic tests are essential

Regulatory Status:
This AI system is for research and educational purposes. It has not been approved by regulatory agencies (FDA, CE, etc.) for clinical diagnostic use.

Professional Review Required:
All findings must be confirmed through standard pathological examination and clinical evaluation by licensed healthcare providers.

TECHNICAL DETAILS

Model Architecture: ResNet50 with Transfer Learning
Training Dataset: Histopathology images (Benign/Malignant classification)
Input Image Size: 224 × 224 pixels
Preprocessing: ImageNet normalization
Explainability Methods: Grad-CAM, SHAP, Integrated Gradients
Software Version: Breast Cancer Detection System v1.0
Processing Timestamp: 2026-02-16 19:29:46

This report was automatically generated by the AI-Assisted Breast Cancer Detection System. For questions or concerns, please consult with your healthcare provider.